

A Real-Time LoRaWAN Tracking System in Railway Tunnels for Increasing Track Maintenance Personnel Safety

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Abstract—Monitoring track maintenance personnel in underground railway tunnels is hampered by the limitations of global positioning system and challenging communication conditions, impacting safety and operational efficiency. This paper presents the design, implementation, and validation of a low-power, Long-Range Wide-Area Network (LoRaWAN)-based tracking system developed to address these concerns. The system employs Received Signal Strength Indicator (RSSI) from LoRaWAN end-nodes for real-time, zone-based localisation and integrates a user-activated Panic Mode for emergency alerts. A comprehensive software architecture utilising a self-hosted ChirpStack Network Server on Arch Linux, Google Cloud Functions for backend logic, and Firebase for real-time data management and web-based dashboard visualisation was implemented using Heltec LoRaWAN hardware. Functional verification in laboratory and campus settings confirmed end-to-end system integrity. Key performance metrics achieved include panic alert and dashboard update latencies under three seconds, and node operational battery life exceeding five hours. Exploratory urban non-line-of-sight tests suggested LoRaWAN connectivity potential up to approximately three kilometres. While this paper establishes a functional proof-of-concept, the paper emphasises that rigorous in-tunnel validation and RSSI calibration are critical for operational deployment. The system demonstrates significant potential for increasing track maintenance personnel safety in underground railway infrastructure.

Keywords—LoRaWAN, LoRa technology, internet of things, real-time location tracking system, monitoring and analytics dashboards, railway track maintenance

I. INTRODUCTION

Efficient public transportation, such as Singapore's Mass Rapid Transit (MRT) system [1], relies on extensive underground tunnel networks, particularly in dense urban areas [2]. Maintaining this underground infrastructure is vital, but it presents safety and operational challenges due to the confined spaces and hazardous conditions [3]. A critical difficulty is the lack of reliable real-time personnel tracking, as conventional methods like global positioning system (GPS) are ineffective underground [4], and short-range wireless technologies often struggle with range and signal penetration through tunnel materials [5]. This technological gap can lead to delayed emergency responses and operational inefficiencies.

Our previous paper [6] introduced a system addressing these tracking needs by employing a hybrid approach that combined Bluetooth Low Energy (BLE) for precise short-range entry/exit zone detection, effective primarily under 100 metres, with LoRa technology for data backhaul. Building upon insights from the initial prototype, this paper presents a distinct and focused investigation into a personnel tracking system based solely on low-power, long-range, wide-area network (LoRaWAN) technology [7]. This current paper details the design, implementation, and validation of a system that utilises Received Signal Strength Indicator (RSSI) from standard LoRaWAN end-nodes to achieve real-time, zone-based localisation over greater distances within tunnels [8], without relying on supplementary short-range beaconing infrastructure. The system incorporates a user-activated Panic Mode and provides real-time visualisation via a web dashboard. The primary objectives were to develop and evaluate this LoRaWAN RSSI-based scalable tracking system for longer-range, continuous zone awareness, enable effective zone localisation using this method, validate the Panic Alert feature, and provide a comprehensive visualisation platform. This paper details these system aspects, implementation specifics, testing, and results.

II. SYSTEM CONSIDERATIONS AND DESIGN

A. Technology Selection: LoRaWAN

The selection of an appropriate tracking technology was a critical first step, requiring careful evaluation against criteria vital for deployment in underground railway tunnels. These criteria included communication range, performance in non-line-of-sight (NLoS) conditions, power consumption for battery-operated personnel devices, achievable localisation accuracy relative to predefined work zones, infrastructure deployment complexity and cost, and overall system scalability. Several candidate technologies were considered. Radio Frequency Identification (RFID) [9] is useful for short-range identification; however, it typically offers insufficient range for continuous personnel tracking over extended tunnel sections. Additionally, its signal penetration through concrete and metal is limited, necessitating a dense reader infrastructure. Bluetooth Low Energy (BLE) [10] is optimised for short-range, low-power communication, but its operational range is significantly reduced in complex environments with obstructions and multipath fading;

providing continuous coverage would require a dense deployment of beacons or gateways. Wi-Fi Positioning Systems (WPS) [11], while potentially leveraging existing infrastructure, often require high access point density for acceptable accuracy and consume considerably more power on client devices. Ultra-Wideband (UWB) [12] offers high-precision localisation but typically requires a dedicated, precisely synchronised infrastructure and often performs best in line-of-sight conditions, with higher costs.

In contrast, LoRaWAN, operating in sub-GHz Industrial, Scientific, and Medical (ISM) bands, emerged as the most suitable technology for the specific objectives of this paper, particularly for achieving broader zone coverage without dense short-range infrastructure. For this paper, the system was configured according to the AS923 regional parameters applicable in Singapore, which typically utilises frequencies within the 920-925 MHz range [13]; this usage is in accordance with the national frequency allocations for Short-Range Devices (SRDs) stipulated in the IMDA's Spectrum Management Handbook [14]. The LoRa physical layer modulation, based on Chirp Spread Spectrum (CSS), provides excellent communication range, good signal penetration through obstacles, and inherent robustness to interference [7]. This significantly reduces the number of gateways needed for wide-area coverage compared to shorter-range technologies. The LoRaWAN network protocol is inherently designed for low power consumption, making it ideal for battery-operated personnel nodes intended for extended use [7]. For this paper, the key hypothesis was that RSSI from LoRaWAN signals could provide sufficient zone-based awareness [8] over considerable distances, offering a potentially simpler infrastructure. Furthermore, as an open standard, LoRaWAN benefits from a wide ecosystem, offering advantages for component sourcing and future scalability. The specific combination of long-range performance from its LoRa physical layer, low transmit power requirements, and CSS robustness made LoRaWAN particularly compelling for the anticipated harsh RF environment within railway tunnels.

B. Overall System Architecture

The implemented system follows a standard LoRaWAN Node-to-Gateway network architecture, where end-nodes (personnel trackers) transmit LoRaWAN messages to gateways, which relay them to a central LoRaWAN Network Server (LNS) [7]. The key hardware components, the personnel node and LoRaWAN gateway, are shown in Fig. 1. The personnel node (Fig. 1a) was prototyped using a Heltec WiFi LoRa32 V3 development board [13]. This board is equipped with an Espressif ESP32-S3 microcontroller, providing sufficient processing power and memory [15], and an integrated Semtech SX1262 LoRa transceiver for AS923 band communication [16]. Its 'PRG' button was repurposed via firmware for Panic Alerts. Each node is powered by a 3.7V Li-ion battery and is housed in a temporary 3D-printed PLA enclosure. The LoRaWAN gateway selected was the Heltec HT-M02 Edge Gateway V2 [17], featuring a Semtech SX1303 and SX1250 baseband chipset (Fig. 1b).

The software infrastructure utilises several key services. A self-hosted ChirpStack V4 LNS [18] on an Arch Linux platform manages LoRaWAN network functions; ChirpStack was selected for its open-source nature, comprehensive

features, and versatility for development and integration. Backend logic is handled by serverless Google Cloud Functions [19]. An HTTP-triggered Cloud Function ingests decoded LoRaWAN application payloads from ChirpStack's HTTPS integration, writing essential device telemetry into the Node Status Collection in Firebase Firestore [20] as shown in Fig. 2. A second, Firestore-triggered Cloud Function, the Zone Determination Function, reads the updated RSSI from the Node Status Collection and applies predefined threshold logic (Table I) to determine the zone ID. This zone ID is then used to update the worker's current zone field within the Personnel Registry Collection (which stores personnel identity linked to their assigned LoRaWAN tag ID) and to update the aggregation of present workers within the Zone Occupancy Collection, as illustrated in Fig. 3. Panic alert events are also logged persistently in a dedicated Alert Log Collection within Firestore, as shown in Fig. 4.

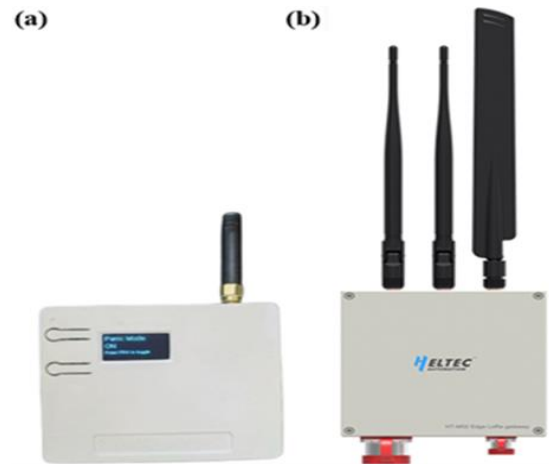


Fig. 1. (a) Personnel node and (b) LoRa edge gateway.

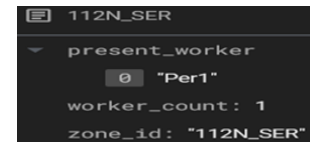


Fig. 2. Node status collection.

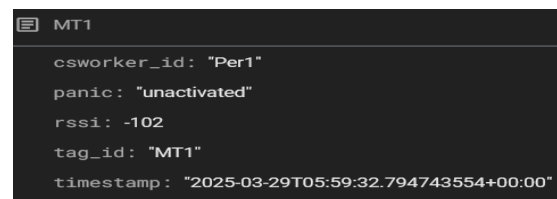


Fig. 3. Zone occupancy collection.

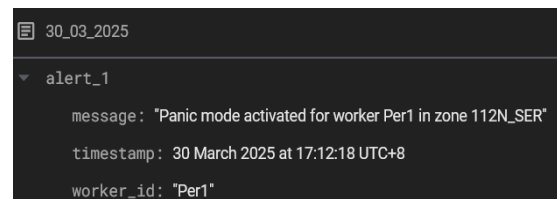


Fig. 4. Alert log collection.

Firebase Firestore was chosen not only for its flexible schema NoSQL document model, suitable for evolving IoT data structures, but critically for its real-time synchronisation

capabilities. This ensures data changes are automatically pushed to connected clients, such as the "RailTime Tracker" web application, without requiring manual refreshes or complex polling mechanisms. The main overview of the "RailTime Tracker" dashboard, displaying the map of monitored zones and general worker presence derived from the Zone Occupancy Collection data, is presented in Fig. 5. The dashboard also provides detailed information for each tracked worker in a tabular format, as seen in Fig. 6. For enhanced interactivity and information retrieval, clicking a specific zone polygon on the main map display reveals a more detailed view or pop-up containing specific information about that zone, such as the Zone ID, Zone Name, current worker count, and a list of present workers within that zone, drawing from the Zone Occupancy Collection data. An example of this detailed zone information display is depicted in Fig. 7.

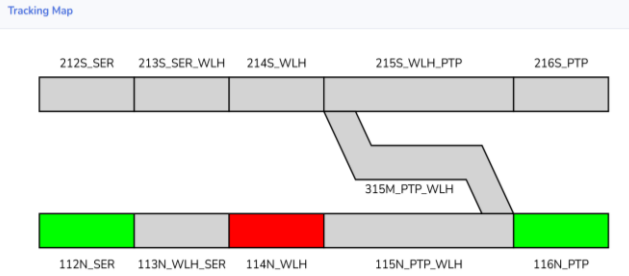


Fig. 5. RailTime tracker dashboard zone mapping visualisation with coloured visuals

csTags

Show 10 entries Search:

tag_id	ti	csworker_id	ti	rss	ti	panic	ti	timestamp	ti
MT1		Per1		-102		unactivated		29 March 2025 at 1:59:32 pm	
MT2		Per2		-64		activated		18 March 2025 at 7:01:24 pm	
MT3		Per3		-34		unactivated		30 March 2025 at 7:01:24 pm	
MT4		Per4		-32		unactivated		30 March 2025 at 7:01:24 pm	

Showing 1 to 4 of 4 entries Previous 1 Next

Fig. 6. Information table.

Station: 112N_SER

Station Details	
Zone Id	112N_SER
Zone Name	Serangeon NB
Worker Count	1

Present Workers

Worker ID: Per1
Worker Details

Fig. 7. Zone information pop-up/view.

C. Core System Functionalities

A primary function of the system is zone-based localisation, visually represented on the dashboard's map, with further zone-specific details such as worker count and present worker IDs available upon user interaction with a zone polygon, as presented in Fig. 7. This method relies on analysing the RSSI of LoRaWAN packets. The underlying principle is that RF signal strength attenuates over distance. This relationship can often be modelled by the log-distance

path loss model [21], where RSSI (in dBm) is expected to decrease logarithmically with distance d from the transmitter:

$$RSSI(d) = A - 10 * n * \log_{10}(d) \quad (1)$$

where A is the received signal strength at a reference distance (usually 1 metre) and n is the path loss exponent, which varies significantly depending on the environment. Raw RSSI data for each tag, along with its timestamp and panic status, is stored in the Node Status Collection (Fig. 2). As previously described, the Zone Determination Function processes this incoming RSSI against predefined thresholds to determine the personnel's current zone. These thresholds, detailed in Table I, are presented in two categories: firstly, 'Conceptual Real-World' estimates, which were developed based on general LoRaWAN propagation characteristics and urban test data proxies, serving as a theoretical baseline for discussion and potential future in-tunnel calibration. Secondly, 'Singapore Institute of Technology (SIT) Campus Grounds' values, which were empirically derived from systematic tests conducted on the university campus, provide a practical demonstration of the zone mapping code's functionality in a controlled, mixed LoS/NLoS environment. It is important to emphasise that while the campus-derived thresholds validate the system's logical capability to differentiate zones based on RSSI, neither set of thresholds in Table I represents validated performance for actual, unique tunnel environments; accurate in-tunnel zone definition and reliable localisation accuracy would necessitate extensive in-situ RF surveys and specific threshold calibration within the target tunnels. This calculated zone information then updates the Personnel Registry Collection with the individual worker's current zone and the Zone Occupancy Collection with aggregated data on worker presence and counts within each specific zone (Fig. 3). This aggregated zone status from the Zone Occupancy Collection directly feeds the visualisations on the dashboard's main map display (Fig. 5) and detailed zone information view (Fig. 7).

TABLE I. RSSI THRESHOLD FOR ZONE MAPPING

Zone ID	Estimated (dBm) - Conceptual Real-World Logic	Measured (dBm) - SIT Campus Grounds (LoS/NLoS)
116N_PTP	> -45	> -40
115N_PTP_WLH	-45 < RSSI ≤ -90	-41 ≤ RSSI < -55
114N_WLH	-90 < RSSI ≤ -120	-55 ≤ RSSI < -65
113N_WLH_SER	-120 < RSSI ≤ -140	-65 ≤ RSSI < -70
112N_SER	< -140	< -70

The Panic Alert system is a critical safety-enhancing feature. Activation of the node's panic button immediately transmits a high-priority LoRaWAN uplink message. This event updates the panic status in the Node Status Collection for the respective worker (Fig. 2) and triggers the creation of a persistent log entry in the Alert Log Collection (Fig. 4), capturing details of the event. The system then dispatches these alerts through multiple channels for timely supervisor awareness. On the "RailTime Tracker" web application dashboard, panic alerts are designed to be highly conspicuous, for instance, by changing the colour of the affected personnel's zone marker on the main map display to a distinct warning

colour and displaying specific alert details. Normal operational status, reflected by periodic uplinks without the panic flag, is also displayed on the dashboard's map, within the detailed information tables, and contributes to the worker count shown in the zone information display. The specific visual outputs of these alert notifications, both on the dashboard and via Telegram, are further detailed and shown in the experimental results section.

D. RF Design Considerations

Conceptual RF design considered standard LoRa antennas and strategic gateway placement within train stations to maximise signal coverage into adjoining tunnel sections. Given the challenging propagation conditions anticipated within tunnels, characterised by signal attenuation due to materials like concrete and rock, as well as complex multipath effects from reflections off tunnel walls and equipment [21], LoRaWAN's Adaptive Data Rate (ADR) mechanism [7] was enabled on the network server. ADR allows the LNS to dynamically command end-nodes to adjust their LoRa modulation parameters (such as spreading factor and transmission power) based on the observed link quality. This optimisation is designed to enhance the robustness of the communication link for nodes located further away or in poor signal conditions, while allowing nodes closer to the gateway to utilise faster, more power-efficient settings, thereby extending the overall battery life. However, the effectiveness of ADR over very long and potentially variable tunnel paths would require in-situ validation. The system's periodic update frequency and the event-driven nature of panic alerts are designed to operate well within the AS923 regional duty cycle regulations [13].

III. SYSTEM TESTING AND VALIDATION

A. Implementation Overview

The development of the personnel tracking node firmware was carried out using the Arduino Integrated Development Environment, specifically targeting the Heltec WiFi LoRa32 V3 board. Key firmware development tasks included selecting and integrating a suitable LoRaWAN library compatible with the AS923 regional parameters used in Singapore, meticulously implementing the Over-the-Air Activation (OTAA) join procedure for secure network registration and defining a concise data payload structure. This payload was designed to efficiently transmit essential status information, most critically the panic flag, which indicates an emergency. Furthermore, software-based debouncing logic was integrated for the repurposed 'PRG' button to prevent false panic alert triggers.

For the network infrastructure, a self-hosted ChirpStack LNS V4 was configured and deployed on an Arch Linux platform. This setup required the installation and configuration of ChirpStack's core components: the Gateway Bridge, responsible for communication with the LoRaWAN gateway; the Network Server, which manages MAC layer functionalities and device sessions; and the Application Server, handling application-level data, device profiles, and integrations. Necessary dependencies, including PostgreSQL for the database and Redis for caching, were also established.

B. Testing Methodology

A multi-faceted testing methodology was employed to verify system functionality and assess performance. Comprehensive functional and integration testing was primarily conducted within a controlled laboratory environment, allowing for the systematic verification of individual component interactions and end-to-end data flow under predictable conditions.

To evaluate the RSSI-based zone mapping logic with empirical data, dedicated tests were performed across the SIT campus grounds. This environment provided a varied setting with both LoS and NLoS paths, allowing for the establishment of the practical 'SIT Campus Grounds' RSSI thresholds presented in Table I. Furthermore, to gain initial insights into LoRaWAN's longer-range NLoS signal propagation characteristics, and in the absence of access to operational tunnels, field tests were conducted in a dense urban residential environment. During these tests, the Heltec HT-M02 V2 gateway was maintained at a fixed, elevated location in a private residential area, while a personnel node was systematically moved to various Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) public positions, reaching distances of approximately three kilometres. The general area and approximate path of the node during these urban NLoS tests are illustrated in Fig. 8.

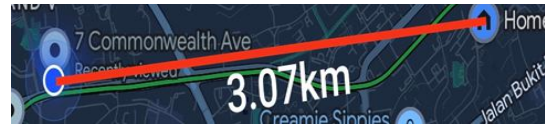


Fig. 8. Urban NLoS test area map.

Verification procedures were designed to assess core system functionalities rigorously. These included confirming the successful completion of the LoRaWAN OTAA join procedure, verifying the consistent transmission of periodic status uplink packets, validating the integrity of the entire data pipeline from node transmission to final display on the "RailTime Tracker" dashboard, and thoroughly testing the panic button activation sequence and subsequent alert propagation. For these performance tests, the end-to-end panic alert latency was defined as the time interval from the physical actuation of the panic button on the personnel node to the corresponding visual alert notification appearing on the web dashboard. The final procedure involved functionally checking the RSSI-based zone mapping logic within the laboratory using the empirically derived 'SIT Campus Grounds' thresholds from Table I.

C. Experimental Results

Functional verification conducted in laboratory and campus settings consistently demonstrated the system's operational integrity. Personnel nodes were reliably powered on, initialised, and successfully joined the LoRaWAN network managed by the ChirpStack LNS. The nodes transmitted periodic status packets at the configured intervals, and the complete end-to-end data flow was robustly verified, with data consistently propagating through the system to update the real-time dashboard views. The integrated panic alert system also performed as expected; manual activation of the node's panic button reliably triggered immediate, high-priority uplink messages. These were correctly processed by

the backend, resulting in appropriate alert notifications. Examples of these alert visualisations are shown in Fig. 9, which depicts both the panic alert indication on the web dashboard (Fig. 9a) and the corresponding notification sent via the 'RailTime' Telegram Bot (Fig. 9b). The RSSI-to-Zone mapping logic, when tested using the 'SIT Campus Grounds' thresholds (Table I), accurately assigned nodes to different predefined zones based on varying RSSI levels. Furthermore, examination of the Firestore database collections confirmed that all node status updates and panic events were being persistently and correctly logged with associated timestamps.

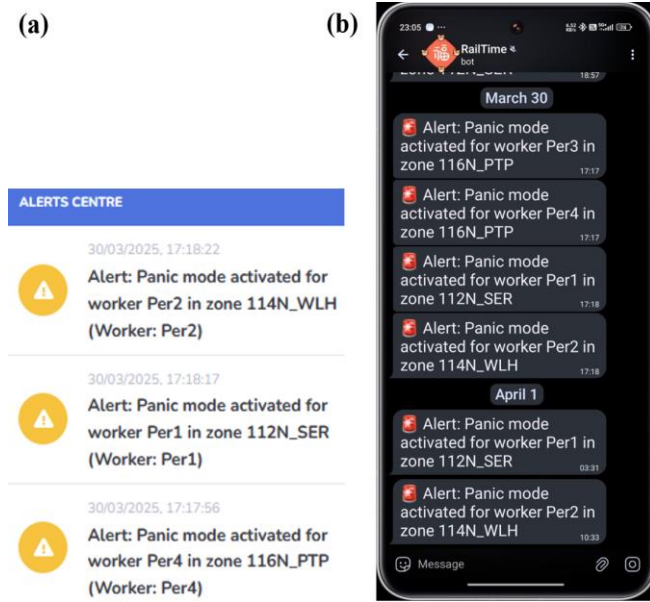


Fig. 9. Panic alert notification system: (a) alert centre on dashboard, (b) emergency alerts notification received via telegram bot.

Performance evaluations, primarily conducted in laboratory settings, yielded promising quantitative results that met or exceeded key design targets. Based on 699 independent test iterations, demonstrating a 100% successful end-to-end transmission rate, the end-to-end panic alert latency averaged 3 seconds, well within the 10-second target. For real-time situational awareness, the dashboard update latency also met its 10-second target, with measured updates consistently occurring in under 3 seconds across all test runs. The personnel node prototype was required to have an operational duration of at least 5 hours for maintenance shifts. To validate this, four prototype nodes were tested concurrently, achieving a cumulative operational duration exceeding 1200 hours. Without employing advanced power-saving modes, each node consistently operated continuously for over 6 hours on a single charge, thereby meeting the design requirement. These key performance metrics are summarised in Table II. The exploratory urban NLoS tests, while not providing formal Packet Delivery Ratio (PDR) metrics, demonstrated that LoRaWAN connectivity was achievable over significant distances (~3 km) with substantial obstructions. The observed RSSI during these tests ranged from -35 dBm (LoS, close range) to -107 dBm (NLoS, ~3 km), providing positive indications for the technology's range capabilities and practical insight into signal levels in complex real-world environments. This observed RSSI range is also included in Table II.

TABLE II. KEY PERFORMANCE METRICS

Performance Metric	Measured Result
Panic Alert Latency (Lab) ^a	3 seconds
Dashboard Update Latency (Lab) ^a	< 3 seconds
Node Operational Duration (Lab) ^b	> 6 hours per node
Urban NLoS RSSI Range Observed (~3 km)	-35dBm to -107dBm

^a Latency metrics established over 699 independent test iterations

^b Based on a cumulative 1200-hour test across four concurrent nodes.

D. Validation Limitations

Despite the positive functional verification and promising performance results, as summarised in Table II, including urban NLoS observations up to approximately 3 km, significant validation limitations persist. The most critical constraint was the lack of access to an operational railway tunnel environment for comprehensive testing and calibration. This directly prevented the empirical validation of key performance metrics under realistic deployment conditions, such as the system's actual communication range, PDR over extended distances within the unique RF propagation environment of tunnels, and the stability and effectiveness of LoRaWAN's ADR mechanism in such conditions.

Similarly, the targeted 90% accuracy for zone identification, which relies on carefully calibrated RSSI thresholds specific to the tunnel environment, could not be empirically tested. While the RSSI-to-zone logic was functionally verified using the 'SIT Campus Grounds' thresholds (Table I), the direct applicability of these campus-derived thresholds to the unique RF propagation characteristics of tunnels is uncertain, and the 'Conceptual Real-World' thresholds presented in Table I remain purely theoretical without empirical tunnel data. Furthermore, the environmental robustness of the temporary 3D-printed PLA node enclosure and a comprehensive ergonomic assessment of its portability, wearability, and panic button accessibility using a finalised design were not performed. These limitations underscore the necessity of future in-situ validation.

IV. CONCLUSIONS

This paper addressed the significant challenge of ensuring personnel safety and maintaining operational oversight during maintenance activities within the GPS-unavailable and communication-hostile environments of underground railway tunnels. It details the design, implementation, and validation of a proof-of-concept LoRaWAN-based personnel tracking system specifically developed to mitigate these challenges by providing real-time zone-based localisation and an emergency alert capability. Building upon insights from our previous paper that utilised a hybrid BLE-LoRa approach for shorter-range detection, this paper focused exclusively on a LoRaWAN solution leveraging RSSI for broader zone awareness without the need for supplementary short-range beaconing.

The developed system architecture successfully integrated Helttec LoRa-enabled personnel nodes, featuring a repurposed panic button, with a LoRaWAN gateway and a self-hosted ChirpStack LNS on an Arch Linux platform. Backend processing was managed by Google Cloud Functions, which

handled data ingestion, zone determination based on RSSI thresholds, as detailed in Table I and alert logic. Firebase Firestore served as the real-time NoSQL database for storing comprehensive operational data, including node status, zone occupancy, and alert logs. A custom "RailTime Tracker" web application provided supervisors with a dynamic dashboard for visualising personnel locations across monitored zones, detailed worker information, specific zone details upon interaction, and critical alert notifications, complemented by Telegram bot alerts. Functional verification, conducted in laboratory and SIT campus settings, confirmed the end-to-end operational integrity of the system. Key performance metrics, summarised in Table II, demonstrated the system's capability to meet design targets under these controlled conditions: panic alert latency averaged 3 ± 1 seconds, dashboard updates occurred in under 3 seconds, and the personnel node operated for over 6 hours on a single battery charge. Exploratory urban NLoS tests indicated LoRaWAN connectivity up to approximately 3 km, with observed RSSI values providing insights into real-world signal behaviour.

While this proof-of-concept robustly demonstrates the functional feasibility of using a LoRaWAN RSSI-based system for enhanced personnel safety and tracking in complex environments, the primary limitation remains the lack of in-tunnel validation. The RSSI thresholds established on campus served to validate the system's logic but cannot be directly extrapolated to the unique RF characteristics of railway tunnels. Consequently, the system's true communication reliability, the achievable accuracy of zone-based localisation, and the performance of ADR within tunnels are yet to be empirically determined. Nevertheless, this paper provides a comprehensive blueprint for such a system and establishes a strong foundation, underscoring that further rigorous in-situ testing and calibration are imperative next steps to transition this promising concept towards a field-deployable safety solution for critical underground railway infrastructure.

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